Ananya Rachuri

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Engineer with 5+ years building data and AI systems that ship and stick in production. Delivered LLM integrations, automation tooling, and enterprise pipelines across healthcare, fintech, and marketing cutting manual ops by 60% on one project alone. Proven expertise in data governance, DQC frameworks. Own work end-to-end, moves fast, and builds things that last.

EDUCATION

Ph.D., Artificial Intelligence | Starting Fall 2026 *School TBD*
Research: AI-Driven Data Quality Automation & Intelligent Analytics Systems

The University of Texas at Dallas | ***Master's in Information Technology Management***
Coursework: Database Systems, Business Intelligence, Machine Learning, Data Governance

SKILLS

Languages & APIs: Python (PySpark, Pandas, NumPy), SQL (Advanced), TypeScript (working proficiency), REST APIs, FastAPI, Git/GitHub
AI / LLM: Claude API, LLM integration, Generative AI pipelines, Agentic automation, Streamlit, scikit-learn, XGBoost, anomaly detection.
Cloud & Data Platforms: Azure (ADF, Synapse, DevOps), AWS (S3, Redshift, Lambda), GCP (BigQuery, Composer), Snowflake, Databricks, Delta Lake
Data Engineering & ETL: Apache Airflow, Kafka, ETL/ELT pipelines, CI/CD (GitHub Actions, Azure DevOps), Docker, Kubernetes
Visualization: Power BI, Tableau, Looker, Google Data Studio
Business & Delivery: Agile/Scrum (JIRA, Confluence), stakeholder management, executive reporting, cross-functional collaboration

WORK EXPERIENCE

TekMind Systems LLC — McKinney, TX
Senior Data Engineer — *Jun 2025 – Present*

- Architected PySpark and Spark SQL feature pipelines in Databricks across 3 concurrent projects, reducing data science prep time by ~35% and accelerating model iteration cycles by 20%, delivering AI-ready datasets to downstream ML teams.
- Designed Delta Lake tables with schema enforcement and ACID-compliant writes; built CI/CD pipelines via GitHub Actions cutting release cycle time by 40% across Dev/QA/Prod.
- Enforced 99.8% SLA adherence using Databricks Jobs and Workflows with proactive alerting for data freshness violations; translated pipeline health metrics into business-readable reports for cross-functional stakeholders.

Cigna Healthcare — Hartford, CT (Remote)
Market Insights Analyst — *Jul 2024 – May 2025*

- Co-built and led "Doctor," an internal automation tool for pipeline monitoring and alerting, reducing manual operations by 60% across VOC ETL pipelines (Qualtrics, Clarabridge, Verint) processing 2M+ records/month owned end-to-end from discovery to deployment.
- Engineered data validation and readiness frameworks achieving 95%+ accuracy across analytics-ready datasets, contributing to a 12% improvement in customer retention forecast accuracy.
- Delivered executive dashboard data and quality-validated reporting inputs for senior stakeholders across a global, matrixed organization, translating complex datasets into clean, decision-ready deliverables.
- Built data cleansing, readiness tracking, and validation frameworks achieving 95%+ accuracy across analytics-ready datasets, contributing to a 12% improvement in customer retention forecast accuracy.

Avis Solutions San Francisco, CA
Data Analyst *Jan 2022 – Dec 2023*

- Owned end-to-end design and iteration of Power BI and Tableau dashboards for Shipium, cutting ad-hoc report requests by 30% through self-service analytics adoption across 500K+ users.
- Designed data ingestion and migration workflows integrating 8 multi-source systems (AWS S3, Redshift, Azure Blob) with DQC controls, achieving 99.1% data accuracy with clean audit trails.
- Translated requirements from analysts, architects, and product owners into scalable Databricks pipeline solutions across Agile sprints, collaborating with cross-functional and Data Domain partners to standardize data processes.

Mindtree India
Junior Data Engineer *Jan 2021 – Dec 2021*

- Developed Python and SQL automation workflows across AWS, GCP, and Azure, executing GAP analysis on 15+ assigned automation opportunities and improving ETL pipeline performance by 25% while reducing manual handling by 3 hours/day.
- Built REST API integrations and SQL pipelines powering Power BI dashboards, conducting data validation, integration testing, and regression testing to maintain pipeline stability across 6 Agile sprints.

PROJECTS

SafeGuard AI — AI Content Moderation System 2026
Python · Claude API · RAG · Flask · Streamlit

- Built an AI system that automatically detects harmful content — hate speech, threats, spam, and misinformation — before it reaches users, correctly classifying 10/10 test cases across 3 verdict types (SAFE/REVIEW/BLOCK) with sub-2-second response time per request and zero false positives on legitimate content including negative reviews, political opinions, and coding humor.
- Designed a policy-grounded moderation engine where content rules live in editable text files, indexed across 13 policy chunks in a vector database — allowing moderation behavior to update instantly without model retraining, reducing policy change time from weeks to minutes.
- Deployed as a live REST API integrated into a 3-platform social media demo (Instagram, Facebook, WhatsApp) processing 10-item batch moderation runs in under 60 seconds — producing 60% SAFE, 10% REVIEW, and 30% BLOCK verdicts on real-world social media content, with a Streamlit dashboard supporting both live and batch moderation modes.

AI-Powered Credit Card Campaign Analytics System 2025
Python · SQLite · SQL (CTEs, Window Functions) · SciPy · Scikit-learn · Claude API · Streamlit · Tableau Public ·

- Built end-to-end analytics system on 50,000 synthetic CRM records — Python ETL → SQLite → SQL analyses (CTEs, window functions: channel ROI, spend lift, segment activation) → logistic regression model achieving 88% precision on churn prediction.
- Integrated Claude API narrative engine auto-generating 4-bullet executive summaries with budget recommendations; AI flags 2% spend anomalies (8–15× normal) with plain-English explanations saved to insights.md.
- Deployed Streamlit chatbot for natural-language campaign queries + 5-sheet Tableau Public dashboard (KPI cards, activation funnel, heatmap, spend lift, interactive Story); full SQLite → Python → Claude API → Streamlit + Tableau pipeline demonstrates LLM-on-BI architecture.

GCP Healthcare Data Pipeline – Python Automation & ETL Platform 2023
Google BigQuery · Apache Airflow · Python · SQL · Git · Cloud-Native Architecture · Raw → Staging → Analytics Mart

- Built automated ETL pipeline (Python + BigQuery) with Airflow DAG orchestration ingesting, staging, and loading healthcare appointment data into analytics marts — processes 50K+ records/run at sub-10 min latency.
- Applied layered data quality checks (PK validation, duplicate detection, date sanity) improving reliability by 30%
- Raw → Staging → Analytics Mart architecture supports scalable no-show rate reporting, mirroring enterprise data lakehouse patterns.

Certifications

Databricks Fundamentals , Generative AI Fundamentals , AWS Academy Graduate - Machine Learning Foundations, AWS Academy Graduate - AWS Academy Data Analytics